

Department of Computer Science
Gujarat University
MSc AI&ML - Semester I
Mathematical Foundation
Coordinate Geometry

1. Let A be $(-3, 5)$ and B be $(5, -10)$. Find (1) the distance AB, (2) the midpoint P of AB. (2)
2. Find the coordinates of the point P which divides the interval A $(-3, -7)$, B $(-1, -4)$ externally in the ratio 4:3. (3)
3. Find the equation of the line parallel to the x-axis which passes through the point where the lines $4x + 3y - 6 = 0$ and $x - 2y - 7 = 0$ meet. (7)

OR

Find the equation of the line which passes through the point $(2, 3)$ and the point of intersection of the lines $3x + 2y = 2$ and $4x + 3y = 7$.

4. Find the equation of the line l through $(1, 3)$ perpendicular to the line $2x + 3y = 12$. Find the equation of the line through $(4, 5)$ parallel to l. (5)
5. What is the distance d of the point P $(-6, -7)$ from the line l with equation $3x + 4y = 11$? (3)
6. Three points A $(3, 6)$, B $(7, 8)$ and C $(5, 2)$ are the vertices of triangle ABC. Find the area of this triangle. (3)
7. Show that the points $(1, 0)$, $(5, 3)$, $(2, 7)$ and $(-2, 4)$ are the vertices of a rhombus. (3)
8. Determine the Cartesian equation of the plane containing the points A $(-1, 2, 5)$, B $(3, 2, 4)$, and C $(-2, -3, 6)$. (9)

